# HEMANTH NOMULA

**AI/ML Engineering | Agentic AI | Backend & Platform Engineering**

*LLM Systems · AI Infrastructure · Applied ML · Enterprise Platforms*

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## SUMMARY

AI/ML and backend/platform engineer building agentic, enterprise-grade AI systems — workspace platforms, procurement intelligence, and finance automation — on Python, FastAPI, PostgreSQL, and Docker. Work spans LLM systems, applied ML evaluation, and production backend architecture, backed by IEEE-submitted research in model robustness and LLM routing.

## SELECTED ENGINEERING

**NexusOS AI — Enterprise AI Workspace**

- Architected a unified AI workspace spanning 15 application modules (conversations, calendar, projects, knowledge, automations) on ~20 core entities — one shared data model and permission boundary across modules — on a Next.js frontend and layered FastAPI service/repository backend with PostgreSQL/Alembic.
- Implemented cookie-based authentication with CSRF protection and Google OAuth connector integration for cross-service account linking.

**VendorLens — AI Procurement & Vendor Intelligence Platform**

- Built a document-ingestion pipeline (PDF/OCR + XLSX/CSV) that reconciles inconsistently formatted vendor quotations into structured pricing, GST, delivery, warranty, and payment fields inside a confidence-scored extraction layer.
- Designed vendor/product matching, landed-cost calculation, and weighted scoring with an approvals workflow and audit trail; integrated Gemini, Gmail, and Google Drive APIs.

**FinPilot AI — Agentic Finance-Control System**

- Architected a deterministic finance-control core (reconciliation, anomaly detection, cash-flow forecasting) where money and approval state are computed deterministically — the LLM only reasons over verified tool outputs and proposes governed actions, never touching ground truth.
- Verified via CI (123/123 tests: 84 core, 21 service, 18 API) plus live Docker smoke tests on a seeded demo dataset; built a full human-in-the-loop approval flow — agent proposes, approver authorizes, execution verified against persisted state and audited.

## SELECTED ML RESEARCH

**Investigating Shortcut Learning in Transformer-Based Fake News Detection** — Submitted, IEEE

- Cross-dataset study (DistilBERT vs. classical TF-IDF + Logistic Regression, Kaggle/ISOT → LIAR): the classical pipeline collapsed from 98.91% in-domain to 48.48% zero-shot accuracy; LIME traced the drop to Reuters-token leakage (83% REAL vs. 1.2% FAKE).

**SmartRouteLLM — Cost-Aware LLM Routing and Optimisation** — Submitted, IEEE

- XGBoost capability-aware router trained on ~33,000 Chatbot Arena samples across 20 models; prompt-embedding features scored 74.89% vs. a 75.58% identity-only baseline, revealing limits of preference-derived routing supervision.

## EXPERIENCE & TRAINING

**Full-Stack Development Intern** — InternsElite, IIT Hyderabad *Dec 2025 – Feb 2026*

- Built a full-stack MERN application with JWT authentication, REST APIs, and persistent CRUD operations.

**AI/ML Training Programme** — Alcheringa, IIT Guwahati *Jan – Apr 2025*

- Completed structured machine learning training, applying core ML concepts through guided implementations.

## TECHNICAL SKILLS

Programming: Python, C++, Java, SQL

ML / AI: PyTorch, Hugging Face Transformers, scikit-learn, XGBoost, LIME, NumPy, Pandas

Backend & Data: FastAPI, PostgreSQL, MongoDB, SQLAlchemy/Alembic, REST APIs, OAuth 2.0

Infrastructure & Tools: Docker, Git/GitHub, Linux, CUDA, Google Colab

Frontend: Next.js, React, JavaScript, TypeScript

## EDUCATION

**B.Tech, Computer Science & Engineering — AI & ML Specialization** *2024 – 2028*

SRM Institute of Science and Technology (SRMIST)

## CERTIFICATIONS

NPTEL Java (Elite Silver) • NPTEL DBMS (Elite) • IBM – Agentic AI for You